Investigating the use of Reinforcement Learning Algorithms in Traffic Environments

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Submission date: Day Month Year

[1]**Disclaimer:** This report is submitted as part requirement for the BSc Degree in Computer Science at UCL. It is substantially the result of my own work except where explicitly indicated in the text. The report may be freely copied and distributed provided the source is explicitly acknowledged.

Abstract

Summarise your report concisely.

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Chapter 1

Introduction

1.1 The Problem

Traffic lights are ubiquitous signalling devices used to control the flow of vehicle and pedestrian traffic, primarily for safety and better regulation of traffic. First introduced in 1868 on Parliament Square in London [Tha17], there have been a wide range of designs created with the most common being a three-colour design – that is green, amber, and red.

The three-colour de jure standard typically uses an internal clock to cycle through and control the duration of each signal phase in a pre-programmed pattern. Whilst this is an obvious approach, the fixed timer takes for granted that traffic demand will remain relatively constant over time, disregarding finer granularity and variations in conditions.

In an endeavour to improve upon this, advanced traffic light systems equipped with sensors are progressively being introduced. These systems dynamically adapt signal timings in response to observed traffic conditions by using rule-based logic and manually designed heuristics. This approach offers improvements in waiting times and vehicle throughput [FHW17], however it relies on predefined rules defined by domain experts that may fail to generalise and perform optimally in all conditions.

1.2 Aims and Objectives

The overall aim of this dissertation will be to perform a detailed analysis and evaluation of how an intelligent agent may be used to hopefully improve upon existing traffic light systems. Rather than explicitly defining behaviour in different traffic scenarios as before, we will observe how optimal decision-making policies may be learnt independently through the use of reinforcement learning algorithms.

!!! BULLET POINT LIST !!!

Our first objective will be to define an environment in which our simulations and experiments will be conducted. Following this we will review the traffic light systems currently in use and then build an abstracted computer representation of the area. The next objective will then be to develop our intelligent agents progressively, going through various reinforcement learning algorithms to see what improvements are made. At each stage, the impact on both pedestrians and vehicles will be considered. We will also consider the impact of changing the road design in itself as a means of making improvements. Our final objective will then be to assess the feasibility of our agents and what factors would need to be considered in a tangible implementation.

Time is a currency that once spent cannot be regained. Traffic lights govern our lives to the extent that saving mere fractions of a second at the individual level can compound to an indescribable amount saved when considering the aggregate. By leveraging intelligent agents, this research hopes to take traffic lights a step further – regardless of how big that step may be.

1.3 Report Overview

We will begin with defining the environment in which we will be conducting our traffic experiments, and then building out this environment using available open source tools and libraries. Whilst accuracy and scope of the environment are important, they will not take precedence over the implementation and analysis of the intelligent agents themselves.

Following this, the context and theory behind reinforcement learning itself will be explored, as well as definitions and terminology that will be used throughout the dissertation. These foundations will be necessary when explaining the motivation behind decisions made later on.

Next we will discuss the creation of an application programming interface that will enable us to connect to our environment for the purposes of observation and control. Based on the default environment and traffic light systems, we will also define a reward which will be used to evaluate performance.

Upon making our environment, we will implement and explore different reinforcement learning algorithms our intelligent agent may use – namely Q-learning and Deep Q-learning. These discussions will be supplemented with the theory regarding how each works, as well as the improvements and limitations proposed by each. Further derivatives of these algorithms and the advantages they bring will also be briefly touched upon, however they will not be implemented.

Continuing from this, a novel approach in which multiple intelligent agents interface together through the use of a network layer will be detailed and implemented. The limitations imposed by such an approach will be considered, as well as how it may be implemented in practice outside of the simulated environment setting.

Finally, after testing and evaluating the different approaches in unforeseen traffic scenarios, we will draw our overall conclusions. Feasibility of such a project will be discussed, as well as interesting technical challenges that were encountered. Future work and limitations to consider will then be commented on.

Chapter 2

Simulated Environment

Before we get to our intelligent agents, we must first construct an environment where the agent can receive input – or state, rather – and perform actions. For the sake of simplicity this environment will be constrained to the streets surrounding University College London's (UCL) Bloomsbury Campus. This area provides scope sufficient for the agents to train well and generalise to other situations.

2.1 OpenStreetMap

OpenStreetMap is a map database maintained by a community of volunteers. Using their data obtained from ariel photo imagery, satellite imagery, geodata sources, as well as manual community additions [HW08], we can extract a portion of UCL's Bloomsbury Campus[1], also complete with data such as traffic lights and crossings.

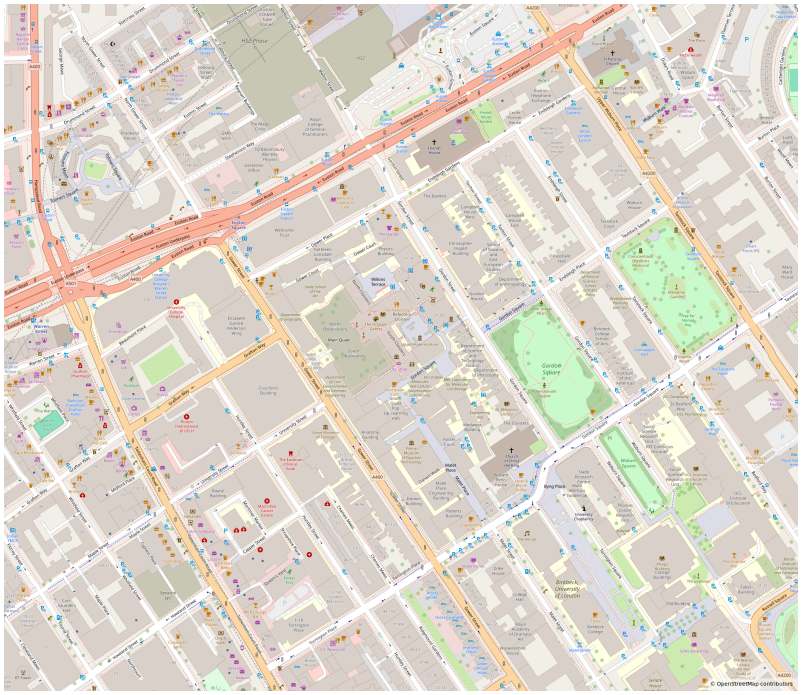


Figure 2.1: Map of UCL's Bloomsbury Campus extracted from OpenStreetMap data.

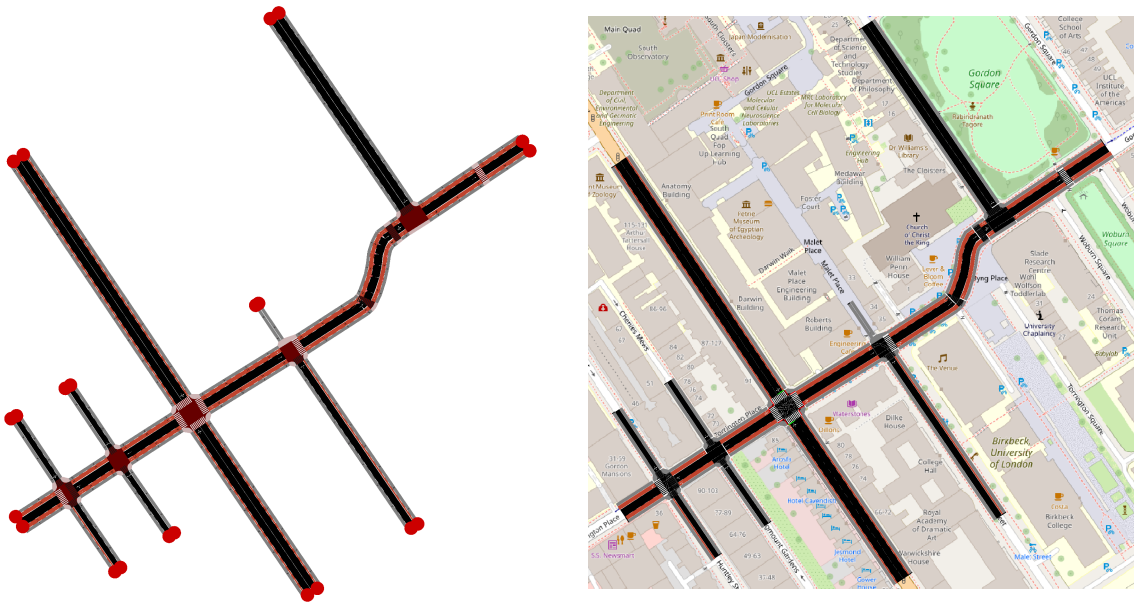
[1]The map centre is: 51.52519°N, 0.12870°W (WGS84).

Although the data from OpenStreetMap provides a lot of useful information, we will supplement it with Google Street View – featured in the Google Maps software – which provides interactive panoramas from positions along the streets. This will allows us to find out more precise information such as lane markings and widths.

2.2 Simulation of Urban MObility

Simulation of Urban MObility (SUMO) is an open source, multi-modal traffic simulation package designed to handle large networks. It allows for intermodal simulation including pedestrians and comes with a large set of tools for scenario creation.

Using the image in Figure 2.1, we can create a network using the *netedit* graphical network editor included in SUMO. Networks are primarily comprised of edges, which is subdivided into lanes, and junctions, which are nodes where the edges meet. Junctions additionally support crossings, lane connections, and traffic lights.



(a) Road network in netedit.

(b) Simulation view in SUMO.

Figure 2.2: Demo simulation environment.

In Figure 2.2a, the red circles and intermediary maroon links indicate junctions, whereas the in between sections are the edges. Each edge can be comprised of a bicycle lane, pedestrian lane, and/or vehicle lane as shown. Additionally, crossings have been added to the junctions, and a configurable traffic light system has been added with the lane connections appropriately adjusted.

Though not visible in the road geometry, lane and junction priorities have also been set, consistent with rules 170–183 of The Highway Code [DfT]. In the UK, 'main' and 'side' roads are relative terms, with the road with priority (e.g. no give-way markings, or priority road signs) functioning as the main road, while the road required to give way is the side road.

2.3 Route Generation

The SUMO installation includes an additional collection of tools for convenience. One such tool is a `randomTrips.py` script written in Python which enables automatic generation of trips and

routes[2] for different origin-destination pairs for a given road network.

We use this script to generate traffic for the following three types of entities, though SUMO provides the flexibility for more:

- Cars
- Bicycles
- Pedestrians

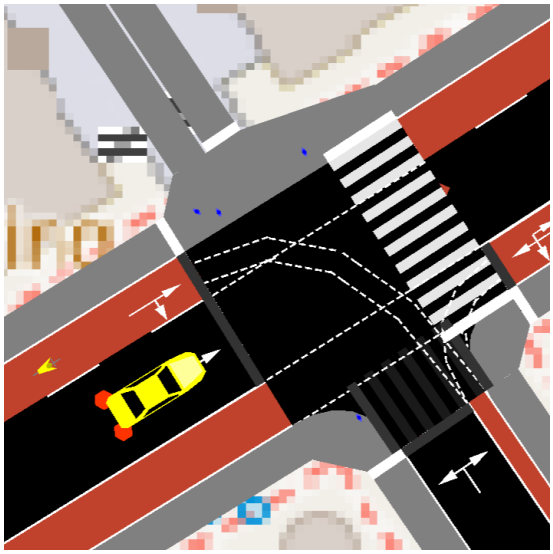


Figure 2.3: Example of car, bicycle, and pedestrian entities.

In Figure 2.3, the car, travelling east, is shown as a yellow rectangle with brake lights; the bicycle, travelling east on the bicycle lane, is depicted as a yellow trapezium-like shape emerging from the left side of the image; and the pedestrians are represented as blue circles.

Route generation is controlled by several parameters, including:

- `seed`: Ensures reproducibility of the random generation process.
- `insertion-density`: Controls the density of generated entities.
- `binomial`: Controls how vehicle insertions are distributed over time by sampling departures from a binomial distribution.
- `fringe-factor`: Ensures that vehicles spawn at the outer edges of the network.
- `random-factor`: Randomizes edge weights to increase variability between generated routes.

There are other obvious parameters omitted which allow you to specify information such as the input network for which the routes are generated, the output trip and route files, the entity type, and the start and end time of entity generation.

[2]N.B. a *trip* defines a random start A and end B, whereas a *route* defines the exact edges to traverse to get from that start A to end B.

2.3.1 Binomial Parameter

The `--binomial` parameter allows vehicle departures to be sampled from a Binomial distribution. Rather than inserting vehicles into the simulation at perfectly regular intervals, stochastic variation is introduced such that some seconds contain more vehicle departures and others fewer.

This is important because real-world traffic arrivals are not perfectly periodic due to the behaviour of drivers as well as variations in the environment. Regular, deterministic insertion can lead to unrealistically synchronised traffic flows.

$$X \sim \text{Binomial}(N, p)$$

$$\mathbb{E}[X] = Np$$

Where Np equals the insertion rate calculated from the `--insertion-density` parameter. Thus, the overall expected value of departures per time step still remains equal to the deterministic insertion rate.

[... image comparing regular interval vs. binomial adjusted insertions]

In traffic theory, Poisson processes are commonly used and preferred over the Binomial distribution to model vehicle arrivals [GS55]. In fact, for sufficiently large N and small p, the binomial distribution approximates a Poisson distribution:

$$\text{Binomial}(N, p) \approx \text{Poisson}(\lambda), \quad \text{where} \quad \lambda = Np$$

The `randomTrips.py` script provides flexibility in the sense that the vehicle traffic itself may be defined and generated using different methods. For example, you may define there to be n fixed 'flows' or routes which traffic may take – some of which may annoyingly be duplicates – or you simply let any of the possible routes for the traffic be arbitrarily, independently chosen each time a new vehicle is inserted.

Unfortunately, whilst there isn't a problem with the Poisson distribution per se, the script is designed such that it is only compatible with the prior traffic generation method and certain parameters. Binomial sampling on the other hand is compatible with the latter method and other parameters which are somewhat more flexible and have less implementation-specific issues. For this reason, it was chosen.

2.4 Summary

Overall, this automated route generation pipeline enables scalable creation of diverse traffic scenarios while maintaining full experimental control.

The main chapters should have a short summary, highlighting what the chapter has covered and any key points that need emphasising.

Chapter 3

Evaluation Framework

To create diverse training and evaluation datasets, we generate multiple route sets using different random seeds and traffic density configurations. The training dataset consists of 1000 route sets per network with baseline density parameters. The evaluation dataset includes several alternative density configurations (e.g., increased and decreased traffic levels), each generated with multiple seeds to ensure statistical diversity.

Routes are stored in a structured directory hierarchy organized by network name, dataset type (train/eval), and random seed. This setup ensures reproducibility and simplifies downstream experimentation.

—

If you have a formal theorem you might try this.

DEFINITION 1 *See definition 1.*

THEOREM 2 *For all* $n \in \mathbb{N},\ 1^n = 1$.

PROOF:

By induction over n. $\square$

Chapter 4

etc.

Chapter 5

Conclusions

5.1 Achievements

Summarise the achievements to confirm the project goals have been met.

5.2 Evaluation

Evaluation of the work (this may be in a separate chapter if there is substantial evaluation).

5.3 Future Work

How the project might be continued, but don't give the impression you ran out of time!

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Appendix A

Other appendices, e.g., code listing

Put your appendix sections here